Inventory Management System

Project Documentation

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Developed as a Flask inventory management project for learning and practice purposes.

Project Description

This project is a web-based Inventory Management System developed with Flask and SQLite. The application allows users to manage company assets, employees, and asset assignments through a clean CRUD interface.

- Employee management
- Asset management
- Asset assignment and return tracking
- CSV import for employees and assets
- Duplicate prevention and validation
- Assignment history tracking
- Asset status management
- Sorting and filtering of records

The application was designed using a layered architecture:

```
routes -> services -> models
```

This structure improves code organization, maintainability, and scalability.

Technologies Used

Category	Technologies
Backend	Python 3, Flask, SQLite3
Frontend	HTML5, CSS3, Jinja2 Templates
Other	CSV processing, SQLite foreign keys, Flask flash messaging

Project Structure

```
final-project/
■
■■■ app.py
■■■ data/
■   ■■■ inventory.db
■■■ models/
■   ■■■ asset.py
■   ■■■ employee.py
■■■ routes/
■   ■■■ assets.py
■   ■■■ employees.py
■   ■■■ assignments.py
■■■ services/
■   ■■■ add_assets.py
■   ■■■ add_employees.py
■   ■■■ assignments.py
■■■ templates/
■■■ static/
■■■ requirements.txt
```

How to Use

1. Start the application

```
python app.py
```

After starting the server, open your browser and access:

```
http://127.0.0.1:5000
```

Asset Management

• Add new assets
• Edit existing assets
• Delete assets
• Import assets from CSV
• View asset assignment history
• Sort assets by category, brand, or status

Employee Management

• Add employees
• Edit employee data
• Delete employees
• Import employees from CSV
• Sort employees by name or department

Assignment Management

• Assign assets to employees
• Return assigned assets
• View active assignments
• View assignment history

CSV Import Formats

Employees CSV Header

```
first_name,last_name,department
```

Assets CSV Header

```
category,brand,serial_number,purchase_date
```

Installation Guide

1	Clone the repository
2	Navigate into the project folder

3	Create a virtual environment
4	Install dependencies
5	Run the application

```
git clone <repository-url>
cd final-proiect

python -m venv venv

# macOS/Linux
source venv/bin/activate

# Windows
venv\Scripts\activate

pip install flask
python app.py
```

Database

The project uses SQLite with relational tables, foreign keys, assignment history tracking, and duplicate prevention mechanisms.

```
data/inventory.db
```

Validation Features

• Duplicate employee prevention
• Duplicate serial number prevention
• CSV header validation
• Invalid row detection
• Date validation
• Required field validation

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